

2.3 SEISMIC PROVISIONS (LRFD 3.10.9, 3.10.9.2 AND 4.7.4)

2.3.1 General

All bridges shall meet the seismic design requirements except the exempted bridges. For exempted bridges, only the minimum bearing support dimensions need to be satisfied as required by **LRFD** 4.7.4.4. Exempted bridges include:

1. Those with design spans less than or equal to 75-feet.
2. Those with simple or continuous span superstructures of any length supported entirely on elastomeric bearings.

For all non-exempt single span bridges, the horizontal design connection force in the restrained direction between the substructure and the superstructure shall be 0.05 times the tributary permanent loads. For all other non-exempt bridges, the horizontal design connection force in the restrained direction between the superstructure and substructure shall be 0.12 times the tributary permanent loads. The acceleration coefficient, A_s , for the state of Florida is less than 0.05. Only the connections between the superstructure and substructure need to be designed for the seismic forces.

2.3.2 Seismic Design for Widening

- A. When seismic design is required for a major widening (see definitions in **SDG** Chapter 7), all new bridge elements must comply with the seismic provisions for new construction.
- B. FDOT will consider seismic provisions for minor widenings on an individual basis.

Modification for Non-Conventional Projects:

Delete **SDG** 2.3.2.B and insert the following:

Do not design minor widenings for seismic provisions unless otherwise required by the RFP.

2.3.3 Lateral Restraint

When lateral restraint of the superstructure is required due to seismic loading, comply with the provisions and requirements of **SDG** Chapter 6, "Lateral Restraint."

2.4 WIND LOADS

2.4.1 Wind Loads on Completed Structures: WL and WS (LRFD 3.8)

- A. Design Wind Speed

Use the design 3-second gust wind speed, V , from Table 2.4.1-1 in lieu of **LRFD** Figure 3.8.1.1.2-1.